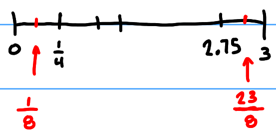
Section 5.2 The Definite Integral

72) Use the Midpoint Rule to estimate $\int_0^3 (x^2 - 3x)dx$ using 12 rectangles.

Find the area under this parabola as x goes from 0 to 3.

$\Delta x = \frac{3-0}{12} = \frac{1}{4}$



$\text{Sum}(\text{seq}((x^2-3x)(\frac{1}{4}), x, \frac{1}{8}, \frac{23}{8}, \frac{1}{4}))$

$\hookrightarrow \boxed{-4.516}$

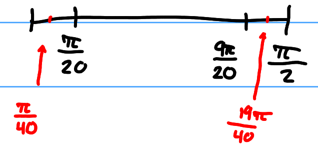
73) It is a fact that $\int_0^{\pi/2} \cos x \, dx = 1$. Find the least number of rectangles required to estimate the definite integral within 0.001 of the actual answer.

estimate within 1.001 and 0.999

try 10 rectangles?

$n = 10$

$\Delta x = \dfrac{\frac{\pi}{2} - 0}{10} = \dfrac{\pi}{20}$

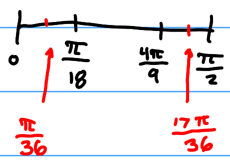


$\text{Sum}\left(\text{seq}\left(\cos x \frac{\pi}{20}, x, \frac{\pi}{40}, \frac{19\pi}{40}, \frac{\pi}{20}\right)\right) = \boxed{1.001}$

But is this the LEAST?

$n = 9$

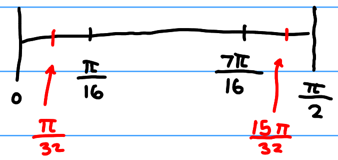
$\Delta x = \dfrac{\frac{\pi}{2} - 0}{9} = \dfrac{\pi}{18}$



$\text{Sum}\left(\text{seq}\left(\cos x \frac{\pi}{18}, x, \frac{\pi}{36}, \frac{17\pi}{36}, \frac{\pi}{18}\right)\right) = \boxed{1.001}$

it's even less rectangles.

$n = 8$, $\Delta x = \dfrac{\frac{\pi}{2} - 0}{8} = \dfrac{\pi}{16}$,



$\text{Sum}\left(\text{seq}\left(\cos x \frac{\pi}{16}, x, \frac{\pi}{32}, \frac{15\pi}{32}, \frac{\pi}{16}\right)\right) = \boxed{0.982}$

off by too much, so 9 was the least

Section 5.3 The Fundamental Theorem of Calculus

74) Evaluate $\int_0^x (t^4 - 3t)\,dt$ and notice that the result is a function of x. Take the derivative of your answer. How does the result compare to the original problem?

$$\int_0^x (t^4 - 3t)\,dt \checkmark$$

$$= \left(\frac{t^5}{5} - \frac{3t^2}{2}\right)\Big|_0^x$$

$$= \frac{x^5}{5} - \frac{3}{2}x^2 - 0$$

$$\frac{d}{dx}\left(\frac{x^5}{5} - \frac{3}{2}x^2\right) = x^4 - 3x \checkmark$$

75) Find the area between the x-axis and the function $f(x)=\dfrac{2}{x^2+1}$ over the interval $[0,1]$.

$$\int_0^1 \frac{2}{x^2+1}\,dx = 2\int_0^1 \frac{1}{x^2+1}\,dx$$

$$= 2\tan^{-1}x\,\Big|_0^1$$

$$= 2\left[\tan^{-1}1 - \tan^{-1}0\right]$$

$$= 2\left[\frac{\pi}{4} - 0\right]$$

$$= \boxed{\frac{\pi}{2}}$$

Section 5.4 Indefinite Integrals

76) Evaluate $\int (\frac{3}{\sqrt{x}} + x^2 + \frac{1}{x})dx$.

$$\int \frac{3x^{-1/2+1}}{\frac{1}{2}} + \frac{x^{2+1}}{3} + \frac{1}{x} \, dx$$

$\frac{1}{x} \downarrow \ln x$

$$= 6x^{1/2} + \frac{x^3}{3} + \ln x + C$$

77) Evaluate $\int \sec x(\sec x + \tan x)\, dx$.

$$\int \left(\sec^2 x + \sec x \cdot \tan x\right) dx$$

$$\boxed{= \tan x + \sec x + C}$$

78) Evaluate $\int \frac{\sin 2x}{\sin x} dx$.

use trig identity

$$\int \frac{2 \cancel{\sin} x \cos x}{\cancel{\sin} x} dx$$

$$2 \int \cos x \, dx$$

$$= \boxed{2 \sin x + C}$$

Section 5.5 The Substitution Rule

79) Evaluate $\int (x+3)\sqrt{x^2+6x+1}\,dx$.

use u-Substitution

$u = x^2 + 6x + 1$

$\frac{du}{dx} = 2x + 6$

$\frac{du}{2} = \frac{2(x+3)\,dx}{2}$

$\frac{1}{2}\,du = x + 3\,dx$

$\frac{1}{2}\int \frac{\sqrt{u}^{+1}}{3/2}\,du$

$\underbrace{\frac{1}{2}\cdot\frac{2}{3}}\,\underbrace{u}^{3/2} + C$

$\frac{1}{3}(x^2 + 6x + 1)^{3/2} + C$

80) Evaluate $\int \frac{x}{\sqrt{1-x^2}}\,dx$ and $\int \frac{x}{\sqrt{1-x^4}}\,dx$.

$$\int \frac{x}{\sqrt{1-x^2}}\,dx, \qquad u = 1 - x^2$$

$$\frac{du}{-2} = \frac{-2x}{-2}\,dx$$

$$-\frac{1}{2}\,du = x\,dx$$

$$-\frac{1}{2}\int \frac{du}{\sqrt{u}}$$

$$-\frac{1}{2}\int \frac{u^{-1/2+1}}{1/2}\,du$$

$$-\frac{1}{2}\cdot\frac{2}{1}\,u^{1/2} + C$$

$$\boxed{-(1-x^2)^{1/2} + C}$$

$$\int \frac{x}{\sqrt{1-x^4}}\,dx \qquad \cancel{u = 1 - x^4}$$

$$\cancel{du = \underline{\underline{-4x^3}}\,dx}$$

Not Involved in original, Not helpful

$$\int \frac{x}{\sqrt{1-(x^2)^2}}\,dx, \qquad u = x^2$$

$$du = 2x\,dx$$

$$\frac{1}{2}\,du = x\,dx$$

$$\frac{1}{2}\int \frac{du}{\sqrt{1-u^2}}$$

$$\frac{1}{2}\sin^{-1}(u) + C$$

$$\boxed{\frac{1}{2}\sin^{-1}(x^2) + C}$$

81) Evaluate $\int x^3\sqrt{x^2+3}\,dx$.

$u = x^2 + 3$

$u - 3 = x^2$

$$\int x^2 \sqrt{x^2+3} \quad x\,dx$$

$du = 2x\,dx$

$\frac{1}{2}\,du = x\,dx$

$$\frac{1}{2}\int (u-3)\sqrt{u}\ du$$

$$\frac{1}{3}\int (u-3)(u)^{1/2}\ du$$

$$\frac{1}{2}\int\left(\frac{u^{3/2\,+1}}{5/2} - \frac{3u^{1/2\,+1}}{3/2}\right) du$$

$$\frac{1}{2}\left[\frac{2}{5}(x^2+3)^{5/2} - 3\,\frac{2}{3}(x^2+3)^{3/2}\right] + C$$

$$\boxed{\frac{1}{5}(x^2+3)^{5/2} - 2(x^2+3)^{3/2} + C}$$

82) Use Riemann sums with $n = 50$ to estimate $\int_0^4 \frac{x}{1+x^2}\,dx$. Next use integration to calculate the exact answer.

$\Delta x = \frac{4-0}{50} = \frac{4}{50} = 0.08$

$\text{Sum}\left(\text{seq}\left(\left(\frac{x}{1+x^2}\right)(0.08),\ x,\ 0.04,\ 3.96,\ 0.08\right)\right)$

$\hookrightarrow \boxed{1.417}$ estimate

$\int_0^4 \frac{x}{1+x^2}\,dx$, $u = 1 + x^2$

$du = 2x\ dx$

$\frac{1}{2}du = x\ dx$

$\frac{1}{2}\int \frac{du}{u}$

$= \frac{1}{2}\ln(u)$

$= \frac{1}{2}\ln(1+x^2)\Big|_0^4$

$\Rightarrow \frac{1}{2}\ln(1+(4)^2) - \frac{1}{2}\ln(1+(0)^2)$

$\Rightarrow \frac{1}{2}\ln(17) - \frac{1}{2}\ln(1)$

$= \frac{1}{2}\ln(17) - 0$

$\frac{1}{2}\ln(17) = \boxed{1.417}$ accurate answer

83) Find the derivative of $g(x)=\int_1^{x^3}\frac{e^t}{t}\,dt$.

$$\frac{d}{dx}\left[\int_1^{x^3}\frac{e^t}{t}\,dt\right]$$

$$=\frac{e^{x^3}}{x^3}\,3x^2$$

$$=\boxed{\frac{3e^{x^3}}{x}}$$

84) Evaluate $\int_0^1 \underbrace{(\sqrt[4]{x}+1)^2}\, dx$

$$(x^{1/4}+1)(x^{1/4}+1)$$

$$= x^{1/2} + 2x^{1/4} + 1$$

$$\int_0^1 \left(\frac{x^{1/2}}{3/2}{}^{+1} + \frac{2x^{1/4}}{5/4}{}^{+1} + 1 \right) dx$$

$$= \frac{2}{3}x^{3/2} + 2\frac{4}{5}x^{5/4} + x \Big|_0^1$$

$$\left[\frac{2}{3}(1)^{3/2} + 2\frac{4}{5}(1)^{5/4} + (1)\right] - \left[\frac{2}{3}(0)^{3/2} + 2\frac{4}{5}(0)^{5/4} + (0)\right]$$

$$\frac{2}{3} + \frac{8}{5} + 1 - 0$$

$$= \boxed{3.267}$$